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Extended Real-Time Monitoring During Multizone Acid Fracturing: Bridging the Data Gap in Non-Instrumented Wells in the Middle East

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Abstract

In multistage fracturing and flowback (FFB) campaigns, real-time downhole monitoring is a crucial enabler of operational efficiency, enhanced reservoir understanding, and improved long-term well performance. In non-instrumented wells, real-time downhole data acquisition (DDA) is especially challenging during longer operations, which extend well beyond the autonomy of typical intervention gauges that communicate through wireless systems in real-time. This paper presents a successful digital strategy that enables continuous DDA in real-time during extended FFB operations lasting over 100 days. Both the telemetry control software and equipment firmware were modified to align the data acquisition plan with the acid FFB stages, enabling optimization of power consumption according to operational phases. In a well without downhole instrumentation available, enabling DDA in real-time allowed for flowback and subsequent pressure buildup (PBU) monitoring after each stimulation cycle. Understanding the results achieved in a previous stimulation stage before starting the next allowed for dynamic decision-making during these complex, multi-zone FFB stages. Furthermore, the well killing process was monitored, guaranteeing a safe POOH after 102 days. The availability of data after stimulation was completed and throughout the fall-off, drawdown, and buildup periods provided a clear picture of operational progress. At the end of the last PBU, the well killing was also monitored, adding value to the decisions made to guarantee the well integrity before safely POOH. In the end, over 800,000 data points were acquired, 50,000 of them in real-time throughout 102 days of operations, a novel result for this setup in the region. It was the first time this modification to the equipment control software and firmware was implemented in the field, and the realtime data acquisition plan was aligned with the operational phases to maximize its utilization through the value-added phases of the job. This led to an extended monitoring of the operations for over 100 days, more than twice the nominal duration of a conventional setup.

Introduction

The equipment was tested in a major offshore oil and gas field located in the Arabian Gulf. Since it was discovered in the late 50's, it remains one of the cornerstone assets of the Middle East production portfolio.

The field is produced from multiple stacked carbonate reservoirs spanning the multiple Cretaceous and Jurassic formations since the 60's. The operator of the field is in a campaign of improving productivity by moving production to different zones, achieved by sidetracking existing wells into new formation layers and doing the needed stimulation processes, depending on the requirements of each geological characteristic, including acid fracturing and acid matrix soaking.

These existing wells usually lack instrumentation, especially on the sidetracked legs, and it is common for stimulation operations to be monitored only through surface parameters. Under these conditions, there is no way to monitor the actual behavior of the fluid downhole during the inflow period immediately after acid pumping is finished.

This Multizone Intelligent Communication is a wireless acoustic telemetry system developed and commercialized in 2013 that enables communication between downhole tools and the surface. As displayed in Figure 1, the acoustic signal transmitted along the string allows real-time data collection from gauge sensors and control of downhole equipment such as samplers, firing heads, downhole valves, packers, production sleeves, and correlation tools. The technology was primarily created for E&A activities worldwide, providing continuous monitoring during all stages of well testing operations—clean-up, main flow, build-up, and well kill. In line with E&A requirements, the system was designed to have an autonomy span of 21 to 28 days under full operational load. Although the system has demonstrated consistent reliability over the years, it became necessary to extend its battery life to meet evolving operational needs and the increasing complexity of projects. Improving autonomy allows for integrating this technology into projects beyond the limited E&A scope, and its applicability to Fracture Flowback (FFB) operations was evaluated during this test.

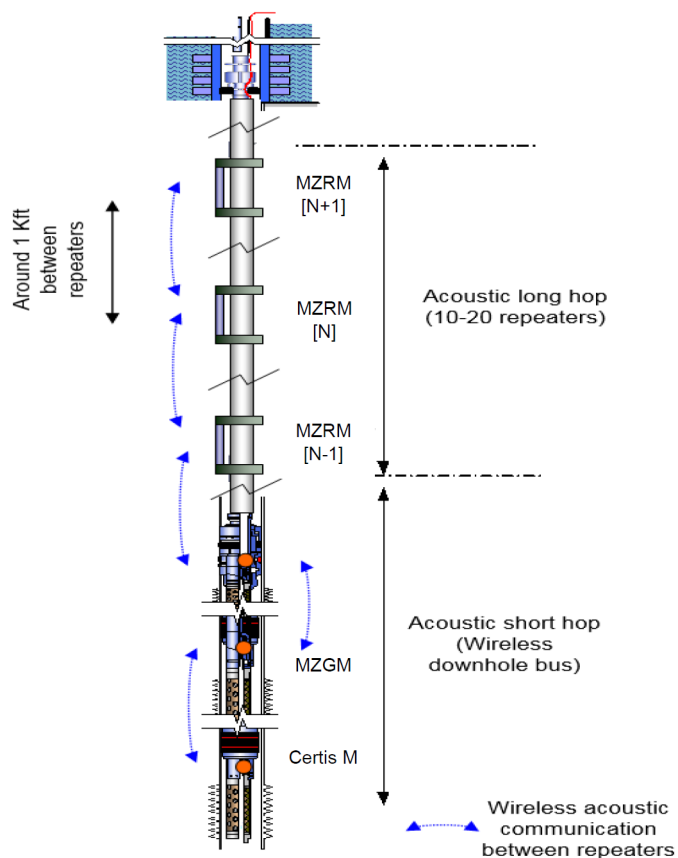


Figure 1—Wireless Acoustic Telemetry Communication System, repeaters, downhole pressure and temperature gauges, and the wireless telemetry activated packer.

Battery life depends entirely on the power consumption of the tools over time. To address the battery limitations, a decision was made to extend autonomy by reducing power to the tools, setting the modems into hibernation mode. Hibernation is a feature that puts the repeater's electronics into a low-power state, with only a real-time clock (RTC) remaining active for a set period. This feature was previously developed and tested, demonstrating its reliability. Still, its use during well testing operations was limited to just a few minutes between queries, with no data available on system performance during longer intervals.

Furthermore, the previous software version only supported a maximum hibernation period of 4 hours (240 minutes) in cyclic mode, while the application was restricted during operation. Based on lessons learned from a prior long monitoring campaign, it was confirmed that a 4-hour hibernation was insufficient to cover typical idle periods such as rigging up equipment, running in hole (RIH), and standby that are common in complex well operations. Another major challenge was limited sensor memory capacity, which was capped at 4 million data sets.

Extending Real-Time Downhole Monitoring Beyond Conventional Limitations

Compared to stimulation services in instrumented wells with permanent downhole gauges, decision-making tends to be less dynamic, and the outcomes of the stimulation are often only understood after the well is completed and instrumented. Adding methods for downhole measurements that enable real-time pressure monitoring during stimulation and flow analysis afterward provides immediate feedback, allowing for dynamic adjustments to the stimulation program, zone by zone. Typically, temporary downhole measurements with real-time transmission are used for exploration and appraisal (E&A) activities. These measurements are inherently designed for the short durations typical of such activities (around 3 to 30 days). They are therefore limited in their application for longer operations, such as multizone frack-flowback and acidizing flowback. Constraints like gauge memory capacity, power consumption of the gauge, power use of the acoustic telemetry system, and battery life limited their performance. The only way to meet these demanding requirements was through careful management of power consumption during both hibernation and data collection. Drawing from lessons learned during a previous extended test, battery performance studies, and a thorough review of the well activity plan, we developed and implemented system enhancements to optimize battery use for both the gauge and modem, thereby extending the operational life of our downhole monitoring tools.

In Fracture Flowback (FFB) operations, the primary objective of the client from a data acquisition standpoint is to monitor pressure buildup and flowback behavior following each stimulation stage. Unlike traditional Drill Stem Testing (DST) operations, FFB does not require continuous real-time monitoring throughout the entire campaign. This distinction in data requirements presented an opportunity to adapt our telemetry technology to better align with the operational goals of FFB campaigns.

To assess the feasibility of deploying the multizone intelligent communication wireless acoustic telemetry system in this context, a detailed review of the well program was performed. This involved identifying and categorizing operational periods into two groups: "idle" intervals where real-time monitoring is not required and "critical" intervals, such as flowback and pressure buildup phases, where real-time data is essential.

This analysis revealed the need to extend the system's hibernation capabilities to avoid unnecessary battery consumption during considered operational idle periods. We also evaluated the technical specifications of the downhole gauges, focusing on scanning rates, as well as the performance of modems and batteries. A key consideration was ensuring that prolonged hibernation would not lead to battery passivation, a condition in which battery capacity is reduced due to extended inactivity. The risk of passivation is exacerbated by the low temperatures observed in the string during the pumping stages.

Field execution and results

Firmware and Software Upgrades

To meet operational needs, an extended modem hibernation feature was created by updating the modem's firmware and software. The upgrade allows the modem to be hibernated for up to 5 days (7200 minutes), 30 times longer than the previous system could support. The telemetry software was also redesigned to support this new capability, giving field engineers more options for hibernation periods and providing tools to manage data acquisition and monitor system status. These significant improvements offer greater flexibility to customize the hibernation feature to match customer requirements.

Having made these changes, a series of intensive tests was conducted to verify that the new hibernation capabilities of the system were functional and operational. Several tests were performed to confirm that all modems connected to the network entered and woke up successfully from hibernation mode after the predetermined periods. Additionally, the battery voltage was monitored throughout all tests to ensure that the batteries could withstand longer periods without passivation or voltage dropping below the cut-off value. Furthermore, the data acquisition strategy was tested to confirm complete synchronization with the operation plan (acid stages, flowback cycles).

Field Execution

The new software and firmware upgrades were carried out during operation by the field engineers. Close communication was maintained among all involved parties: the stimulation team, surface team, client stakeholder, and reservoir engineers to ensure clear alignment on the hibernation program and the testing plan. This collaboration ensured the proper scheduling of hibernation during idle times to conserve battery life when data collection was not needed.

After each hibernation period was concluded, the field crew could track system performance by monitoring how well each modem was performing on the string, utilizing the "Modem Health" feature. The software offered an accurate estimate of available battery autonomy and monitored the battery voltage of the modems. This also enabled the team to monitor battery trends and verify that all tools ran above the safety cutoff voltage. This proactive management, enabled by the upgrade, was pivotal for achieving the remarkable operation time. By tuning these parameters, we sought to enhance the overall autonomy and reliability of the system, enabling its successful implementation in long-duration FFB operation while maintaining data integrity during the critical monitoring periods.

Figure 2 shows the distribution of the duration (in days) of each one of the stimulated zones. It is noticeable that each one of the intervened zones has an average duration of a complete E&A operation.

By maintaining the battery's life and ensuring optimal performance during the entire test, five zones were tested effectively. Throughout the full test, real-time data was transmitted, enabling the customer to understand the performance and the behavior of each zone. The downhole data and interpretation provided to the customer enabled well-informed and timely decision-making during the testing operation.

Each of these zones underwent matrix stimulation aimed at increasing the reservoir's permeability and boosting its overall productivity. After each stimulation operation, the downhole data collected became a crucial asset for assessing the effectiveness of the stimulation. This data offered a clear and accurate measure of how each job contributed to the reservoir's performance. The prompt and precise delivery of this data ensured that the customer had the necessary tools to make real-time decisions, especially when planning the next steps in the well's flowback operations. This smooth integration of technology and data-driven decision-making allowed for enhanced project efficiency and results.

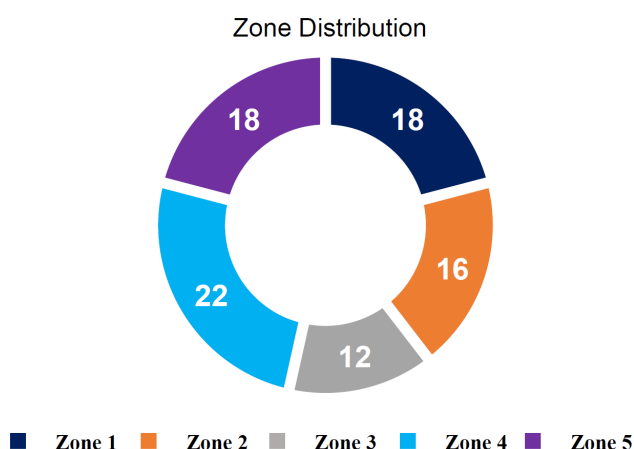


Figure 2—Distribution of data (in days) for each of the stimulated zones within the field operation

Data Acquisition results

Acid treatment control enhances safety margins and reduces overall operation time. Each of these applications demonstrated the importance of having access to real-time downhole data during extended operations. In the next section, the mentioned applications will be detailed individually, highlighting the efficiency improvements from real-time data and lessons that can be applied to similar projects. The implementation of the Muzic telemetry upgrade for this job provided tangible value during the main stages of the test. From better

Throughout the entire operation, Communication was established with the top node 99% of the time. Over 4000 queries were sent, and more than 56900 bytes of data were received at the surface. Each query took (on average) less than a minute to return data to the surface. Communication efficiency with the downhole gauges was efficiently maintained during the active periods, and a total of 50,000 pressure points were received in real time from 8 different gauges.

At the end of the operation, the string was POOH, and the memory of the gauges was read, yielding over 800,000 datapoints from the same eight gauges (pressure and temperature). Figure 3 shows the entire job history of the operation, including zone data and well kill data.

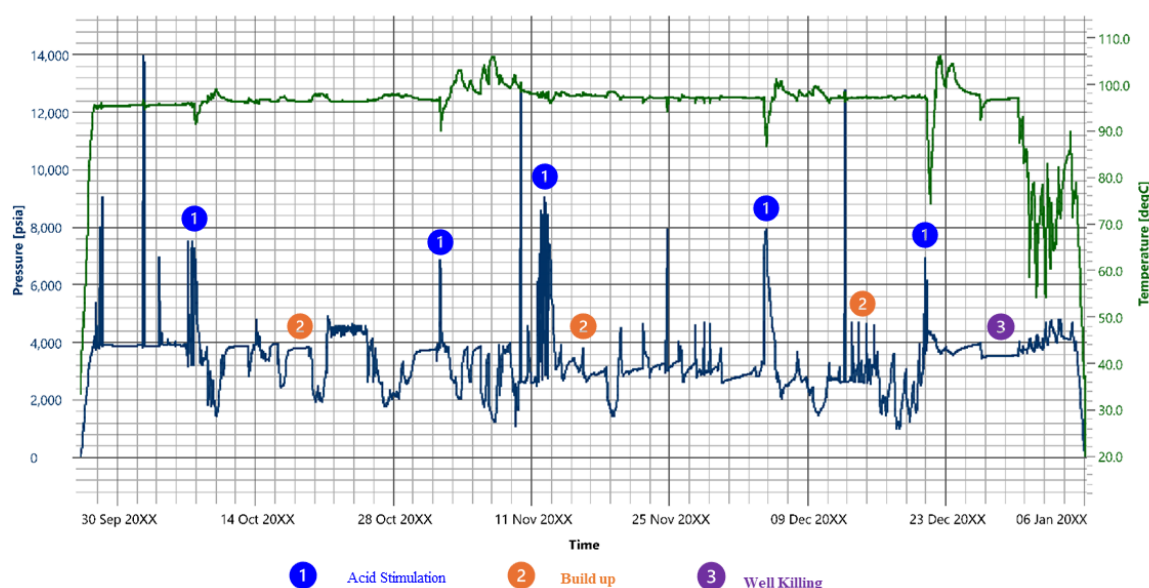


Figure 3—Downhole data (BHP and BHT) acquired during the operation - Readings from Gauge 1 of 8

Efficiency Gains

Acid Stimulation Optimization

During matrix acidizing marked as 1 in Figure 3, live downhole pressure and temperature readings were found to be critical in guiding the pumping strategy. By tracking downhole pressure gauge data in real time, bottomhole pressure was maintained safely below fracture limits while pump rates were increased to maximize wormhole penetration and reduce skin. Stimulation effectiveness was thereby improved, and clear feedback on diverter volumes was obtained, enabling fluids to be distributed more evenly across the target interval. As shown in Figure 4, pressure and temperature data from the gauges were also utilized to fine-tune the corrosion inhibitor package to match actual well conditions rather than relying solely on pre-job estimates. Reliance on conservative pre-job designs was reduced, and adjustments to changes in the well's response were made in real time.

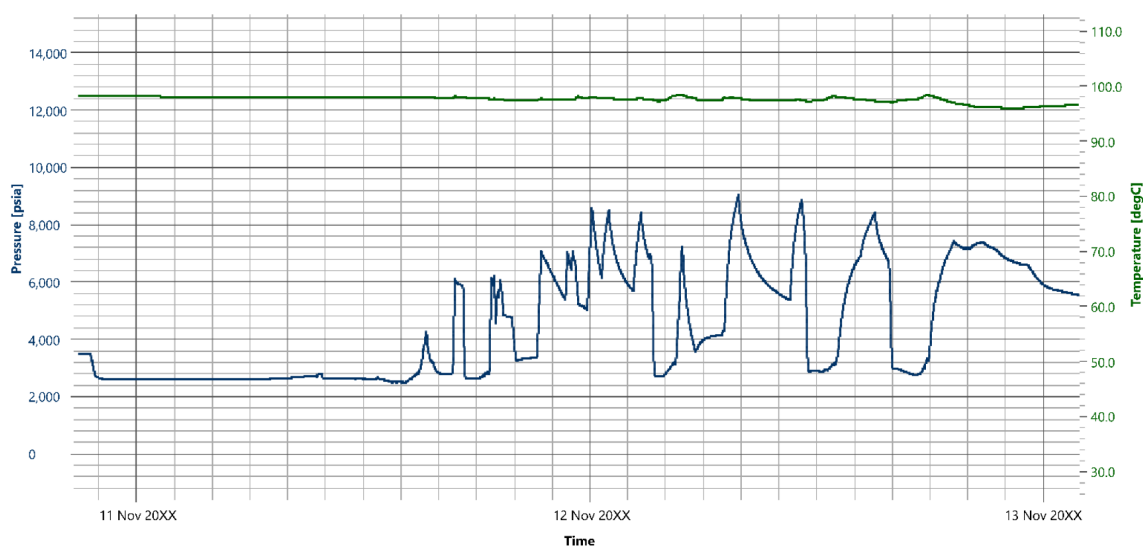


Figure 4—Zone 3 Acid Stimulation

Due to the feasibility of the timely tuning of acid parameters, each stage was executed with higher efficiency and treatment quality, resulting in a quantifiable improvement in post-job productivity. The ability to monitor these parameters continuously also ensured that any deviation from the planned zone treatment design was identified and addressed on the spot. Thus, the acid treatment jobs were carried out with both improved operational control and enhanced reservoir response.

Optimizing Pressure Buildup Duration

Label 2 in Figure 3 shows that post-stimulation pressure buildup periods were optimized and reduced based on real-time downhole pressure data. Rather than using predetermined periods, pressure stabilization was actively monitored, and the buildup period was terminated when the required reservoir data were obtained (Figure 5). Relying on real-time data reduced nonproductive time, preserved rig time, and ensured the collection of reliable reservoir evaluation data.

By eliminating unnecessary waiting times between stages, the customer was able to proceed with the campaign faster without compromising downhole data quality. Additionally, the customer had the ability to modify the buildup periods based on the actual well conditions, which led to lowering operational costs and releasing unnecessary resources required for later stages. Thus, for every phase, the buildup period was reduced significantly while still meeting the operational objective. The use of live data also ensured that each buildup was not terminated until reservoir behavior had been fully captured. As a result, zone evaluation became both more accurate and time-efficient.

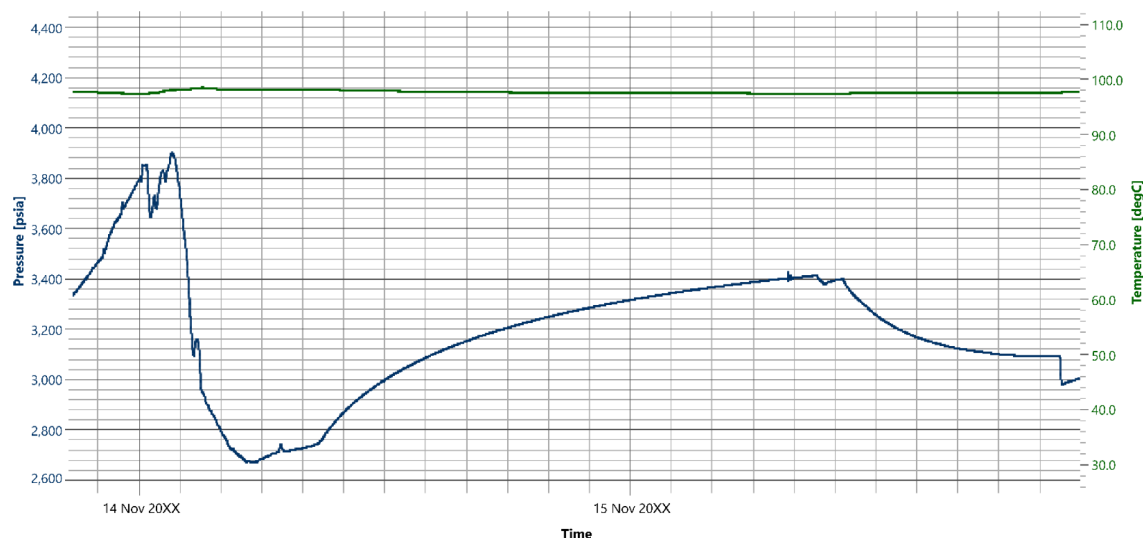


Figure 5—Zone 3 Buildup

Well Killing Decision Support

Marked as 3 in Figure 3, real-time monitoring of downhole pressure and temperature served as key safety measures during well killing activity. Before unsetting the packer and pulling the string out of the hole, the customer had to confirm that the hydrostatic barrier was in place, which was possible with accessibility to real-time downhole data. Furthermore, having real-time downhole data provides flexibility to adjust the kill fluid weight based on actual downhole conditions, rather than assumptions. Additionally, real-time data contributed to reducing human errors, minimizing unnecessary delays in the ability to make timely decisions, resulting in the well killing operation being conducted under safe and controlled conditions. As shown in Figure 6, any abnormal pressure changes would have been detected early through real-time trend monitoring, preventing potential well control problems.

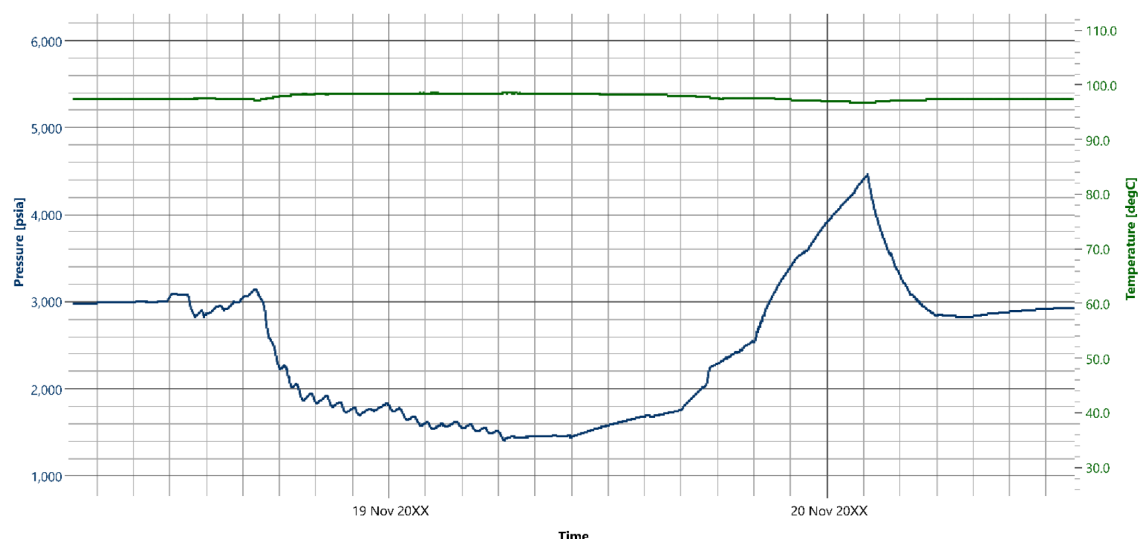


Figure 6—Well Killing monitoring - real-time data allowed for a confident decision on when it was safe to POOH.

As a result, the well killing operation became more efficient, with decisions based on live and reliable data instead of estimates. To Elaborate, the long well monitoring periods post well killing were significantly reduced due to the availability of continuous downhole data. Allowing the customer to proceed to the next

operational step faster. As a consequence, well control assurance was achieved more quickly, leading to a safer and more efficient completion of the operation.

Hibernation Performance Validation and Autonomy Margin

The implementation of the long-term monitoring in this campaign provided a valuable insight into the practical performance of the upgraded hibernation feature under real operation conditions. As shown in Figure 7, by the end of the operation, more than 30% of the battery capacity remained, which is equivalent to an estimated additional 30 days of operations, considering the environment the equipment was used in.

Description		Modem Status			
Direction	Modem Id	Name	Reception Time	Battery (Volt)	Estimate Battery (%)
Up	27	MZHTR-C 898		6.6	23
Up	26	MZHTR-C 902		6.5	50
Up	25	MZHTR-C 825		6.4	51
Up	24	MZHTR-C 899		6.6	36
Up	23	MZHTR-C 772		6.5	46
Up	22	MZHTR-C 901		6.4	51
Up	21	MZHTR-C 917		6.5	47
Up	20	MZHTR-C 911		6.5	44
Up	19	MZHTR-C 900		6.6	47
Up	18	MZHTR-C 904		6.6	48
Up	17	MZHTR-C 896		6.5	42
Up	16	MZHTR-C 920		6.7	31
Up	15	MZHTR-C 786		6.6	48
Up	13	MZHTR-C 762		6.5	45
Up	12	MZHTR-C 921		6.6	29
Up	11	MZHTR-C 767		6.6	47
Up	10	MZHTR-C 903		6.6	37
Up	9	MZHTR-C 919		6.6	32
Up	8	MZHTR-C 910		6.6	37
Up	7	MZHTR-C 859		6.5	38
Up	6	MZHTR-C 897		6.4	37
Up	5	MZHTR-C 918		6.4	38
Up	3	MZHTR-C 869		6.3	35
Up	2	MZHTR-C 824		6.2	22

Figure 7—Remaining Battery Percentage read once the tools were POOH at the end of the operation - indication that the remaining battery capacity would allow for a continuation of the communication for another 30 days.

This result confirmed that the extended hibernation intervals not only met but exceeded the autonomy requirements for the campaign. It also gave all involved parties confidence to plan for operations expected to last more than 100 days, leading to increased opportunities for downhole monitoring in long-term interventions planned in non-instrumented wells.

Conclusion

The extended deployment of the upgraded acoustic telemetry system in this multizone fracture flowback campaign demonstrated that long-term, real-time downhole monitoring is achievable in non-instrumented wells through hardware improvements, firmware updates, and a tailored data acquisition plan. By aligning the data collection schedule with actual operational stages and fully utilizing extended hibernation features, battery life was maintained for over 100 days, more than doubling the usual duration with standard setups.

From an operational viewpoint, the benefits of live downhole data were evident at various stages:

- During matrix acidizing, real-time downhole pressure gauge data enabled pumping rates to be optimized while keeping bottomhole pressure below fracture limits. This immediate feedback also clearly illustrated how diverter volumes influenced fluid placement, resulting in a more uniform distribution along the interval. Temperature readings from the gauges allowed for instant adjustments to the corrosion inhibitor blend.
- During well killing, live pressure and temperature trends validated barrier stability and helped precisely adjust the kill fluid density. For post-stimulation buildups, pressure stabilization was

tracked in real time, facilitating early completion of the buildup phase without compromising reservoir data quality, which saved hours- and in some cases, days- before progressing to the next stimulation step.

From a technical perspective, extending hibernation durations was essential for increasing system autonomy. Issues such as battery passivation and memory saturation were managed through pre-job simulations, in-field validation, and continuous system health monitoring. Communication reliability remained high, with over 50,000 data points collected in real time and more than 800,000 retrieved from memory after recovery. The development team observed improvements, now being implemented and tested in the field, to further enhance system performance during future long-duration deployments.

Overall, the campaign demonstrated how adapting telemetry operations to real-time data needs can bridge the instrumentation gap in complex interventions. The lessons learned provide a proven framework for employing extended-duration real-time monitoring in similar long-term test projects, supporting quicker decisions, safer execution, and more informed reservoir management. The results on the performance of the modified system lead to a belief that the utilization of the assets previously developed for E&A can also support real-time downhole monitoring conditions for multiple other operations, including those occurring over an extended period.

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Nomenclature

<i>DDA</i>	downhole data acquisition
<i>DST</i>	Drill Stem Testing
<i>E&A</i>	exploration and appraisal
<i>FFB</i>	fracturing and flowback
<i>PBU</i>	Pressure Buildup
<i>POOH</i>	Pull out of the Hole
<i>RIH</i>	Run in Hole
<i>RTC</i>	Real Time Clock

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